



United International University (UIU)  
Dept. of Computer Science & Engineering (CSE)

Class Test, Trimester: Summer 2025  
Course Code: CSE-1325, Course Title: Digital Logic Design  
Total Marks: 20, Duration: 40 Mins

**Q1. Smart Traffic Signal Controller (T Flip-Flop Sequential Circuit)** [10]

You are designing a smart traffic signal controller for a city intersection. The system uses two T flip-flops ( $T_2, T_1$ ) to represent its internal control state and one external input  $X$  indicating traffic density:  $X = 0$ : Light traffic (normal mode),  $X = 1$ : Heavy traffic (adaptive mode).

**Behavior** — When  $X = 0$ : The controller drives the next state to 00 (base configuration) on the next clock, regardless of the current state. When  $X = 1$ : The controller cycles through states to adapt timing:  $00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00$  (repeat).

**Tasks:**

- Draw the state diagram (include  $X$  on arcs).
- Construct the state table with Present State ( ~~$T_2, T_1$~~  <sup>(Q2, Q1)</sup>), Input ( $X$ ), Next State, and required flip-flop inputs ( $T_2, T_1$ ) <sup>(Q2+, Q1+)</sup>.
- Derive minimized Boolean equations for  $T_2$  and  $T_1$ .
- Draw the circuit diagram using the equations, T flip-flops, and standard gates (assume synchronous operation)

**Q2. Synchronous Counter Design (D Flip-Flops)** [10]

A company has called 6 candidates assigning serial numbers from 1 to 6 for an interview. On the day of the interview, the board wants to call the candidates in the following arbitrary order:  $4 \rightarrow 1 \rightarrow 6 \rightarrow 2 \rightarrow 5 \rightarrow 3 \rightarrow 4 \rightarrow \dots$

Design a synchronous counter using **D flip-flops** and basic gates to generate this sequence. For any unwanted/unused number, assume the next number is 1.

You must show: **(i)** the state table, **(ii)** the optimized input functions, and **(iii)** the circuit diagram.

Q3. As we wrap up the trimester, here is a token of gratitude of 2 marks bonus. I wish you all the best. [02]